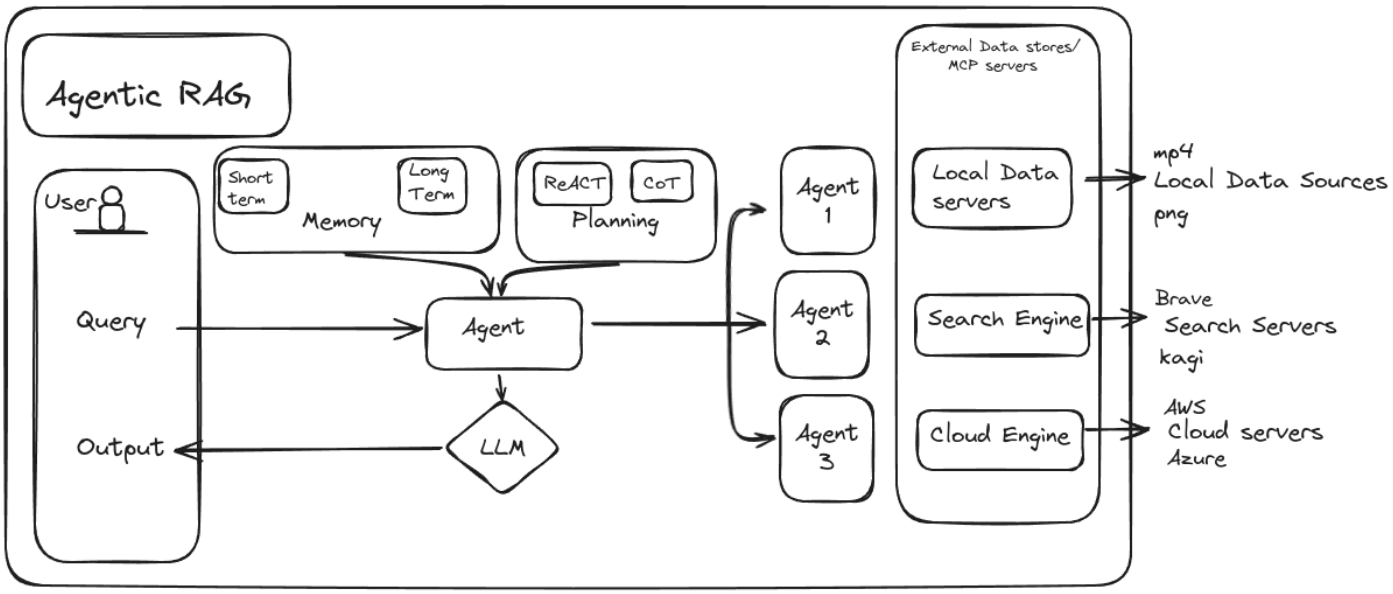
Hey, Thank you for applying!

Below is the Task deliverable that you need to submit to move to the next round, for the Role of AI Engineer at 6th-sense AI.

Internship project brief — Agentic RAG

Your Task : Build This



Key note (read first): The *full* Agentic RAG pipeline (planner → multiple agents → retrieval + web search → verifier + tool/compute → LLM synthesis + memory + audit log) is the ideal target and will get top marks. **However** any *partial* implementation that works end-to-end (even a simple RAG) is fully acceptable and encouraged — deliver something reliable rather than an incomplete, brittle “everything” demo.

Use-case

You may choose any use-case for the project.

Simple example use-case (keep it tiny)

Below is an **example** use-case you may use.

Invoice → PO matcher (recommended) modify for complete project implementation
User asks: “Why was invoice INV-123 flagged?”

System goal: explain *why* the system flagged an invoice by returning a short human-readable explanation, the evidence used (PO, goods receipt, invoice lines), a match score, verifier confidence, and an audit log of planner/agent actions. It should also support a follow-up, e.g., “Approve it” (Provide a mock output response).

Minimal deliverable (required)

1. **Code Repo** (runnable, with setup steps).
2. **README**: how to run, architecture diagram mapping components to your code, and known limitations.
3. **Demo**: one 3–6 minute video **OR** 6–10 annotated screenshots showing (a) architecture diagram, (b) 2–3 demo queries, (c) retrieved evidence and agent outputs, and (d) audit log snippet.
4. **Short doc** : chosen use-case, what was implemented vs omitted, Limitations ,recommended next steps. Give your process of thinking and building.

Extra (nice to have)

- Prompt templates for planner / verifier / synthesizer.
- Unit tests for each step
- Small dataset and test queries.
- Sample audit log in JSON lines format.
- Short-term memory demo: ask a follow-up (e.g., “Okay, approve it” after “Why was it flagged?”).

Tips & pitfalls (practical)

- **Start small & reliable:** a simple, well-explainable retriever + LLM with sources is better than a fragile multi-agent pipeline.
- **Always show provenance:** include source names and snippet offsets in the LLM answer — this demonstrates grounding.
- **Keep planner simple:** rule-based planner is fine for the demo (e.g., “if invoice contains PO# → call PO-matcher; else → only retrieve”).

- **Set thresholds:** verifier confidence thresholds determine whether to auto-respond or escalate to human. Tune on your samples.
- **Parallel calls need merging logic:** when local and web results disagree, show both and let verifier highlight conflicts.
- **Protect data:** redact PII in demo data; do not expose real vendor/customer info.
- **Log EVERYTHING:** planner choices and agent outputs — these make the demo trustworthy and debuggable.
- **Test edge cases:** missing PO, partial matches, OCR noise — show how system fails gracefully.
- **Human-in-the-loop:** for any action with monetary or employee impact, require manual confirmation in the demo.
- **Document limitations:** be explicit about what you didn't implement and why — that's a positive.

Submission

Expecting project submissions within 4 days time (1st sept)
Submit everything in a Single DOC to this Link.

https://www.dropbox.com/request/W5BYlqoddFvlezWE4KEA

CONTAINING:

Code repo Link (runnable, with setup steps).
README: how to run, architecture diagram mapping components to your code, and known limitations.
Demo: one 3–6 minute video **OR** 6–10 annotated screenshots showing (a) architecture diagram, (b) 2–3 demo queries, (c) retrieved evidence and agent outputs, and (d) audit log snippet.
Short doc : chosen use-case, what was implemented vs omitted, Limitations ,recommended next steps. Give your process of thinking and building.